Program : **Diploma in Communication and Computer Networking**	
Course Code: **4329**	Course Title: **Database Lab**
Semester : **4**	Credits: **1.5**
Course Category: **Programme core course**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- Practice different DDL,DML and DCL commands
- Apply Database connectivity via programming languages like JAVA and PHP
- To Understand Database Design

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Prepare a database schema for a real world problem using standard approaches	10	Applying
CO2	Construct queries using SQL for database creation, interaction, modification, and updation.	8	Applying
CO3	Write procedures, functions, and control structures using PL/SQL	9	Applying
CO4	Demonstrate database applications using front-end tools and back-end DBMS	10	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			1
CO2	3			3			1
CO3	3			3			1
CO4	3			3	2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Sample Core Problem: Develop the academic functioning of a college having five entities viz.

Department, Faculty, Student, Course, and Hostel.

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Prepare a database schema for a real world problem using standard approaches**		
M1.01	Analyse a real problem situation/application and represent it using an ER Diagram. Create tables as per ER Diagram and insert relevant data.	4	Applying
M1.02	Create various views of the table	3	Applying
M1.03	Delete unwanted tuples from tables. Modify the schema of one or more tables to incorporate a change	3	Applying
CO2	**Construct queries using SQL for database creation, interaction, modification, and updation**		
M2.01	Create one or two tables. Bring in all possible constraints like NOT NULL, DEFAULT, CHECK, PRIMARY KEY, UNIQUE etc.	4	Applying
M2.02	Write various SELECT clauses to generate different types of listing of data. (Use WHERE, HAVING, DISTINCT, GROUP BY and ORDER BY clauses and subqueries)	4	Applying
	Series Test– I	2	
CO3	**Write procedures, functions, and control structures using PL/SQL**		
M3.01	Creation of Procedures, Triggers and Functions	3	Applying
M3.02	Implementation of various aggregate functions in SQL	3	Applying
M3.03	Implementation of various control structures like IF-THEN, IF-THEN-ELSE, IF-THEN-ELSIF, CASE, WHILE using PL/SQL	3	Applying
CO4	**Demonstrate database applications using front-end tools and back-end DBMS**		
M4.01	Develop the miniature form of application that connects to the database using JDBC/ODBC	3	Applying
M4.02	Design and develop a form to enter data into tables using JAVA/PHP	3	Applying
M4.03	Develop a Java/PHP program to access the table and generate a report.	4	Applying
	Series Test– II	2	

	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.
Minimum of 6 experiments (excluding open ended experiments/microprojects) shall be performed.
Compulsory for CIA
The CIA shall be arranged in third semester by the faculty in charge.The ESE need to be conducted at the end of the third semester.

Text / Reference

T/R	Book Title/Author
T1	Elmasri R. and S. Navathe, Database Systems: Models, Languages, Design and Application Programming, Pearson Education, 2013
T2	Sliberschatz A., H. F. Korth and S. Sudarshan, Database System Concepts, 6/e, McGraw Hill, 2011.
R1	Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015
R2	NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases and Big Data), Wiley, 2018
R3	Database Management System Oracle SQl and PL/SQL, Pranab Kumar das Gupta, PHI, 2009 Print

Online Resources

Sl.No	Website Link
1	http://www.sql-tutorial.net/
2	https://www.w3schools.com/sql/
3	https://sqlbolt.com/